

Explain the health investment function and home good function in the light of Grossman Model?

Ug 22al

An increment to capital stock such as health is called an investment. During Each period a consumer produces an investment in health I .

I is produced by time spend producing health T_H and market health inputs M .

Home good B as baking bread market goods might include flour yeast, kitchen appliances, gas etc.

Thus the consumer using money to buy health care inputs (M) or home good inputs (X). He is using leisure time either health care (T_H) or producing home good (T_B).

Using functional Notation :

$$I = I(M, T_H)$$

$$B = B(X, T_B)$$

These functions indicate that increased amount of M, T_H increase I and increased amount of X, T_B increase B . ✓

In this model Consumers ultimate resource is his own time. Treat each period of analysis as being a year in length and assume 365 days available in the year. To buy market goods M or X , he must trade off some of this time for income that is he must work at job. (T_W). We realize that some of his time during each year will be taken over by ill health (T_L). Thus we account for his total time in the following manner :

$$\text{Total time } = T = 365 \text{ days} = T_H (\text{Improving health}) + T_B (\text{producing home good}) + T_L (\text{lost to illness}) + T_W (\text{working})$$

Recall that his leisure time is spent either improving health or producing home goods.



How would you explain the potential use of a person's time by the labor-leisure trade-off.

Ans:-

The potential uses of a consumer's time are illustrated by the labor-leisure trade-off. Our variation on this analysis also helps illustrate the investment aspects of health demand.

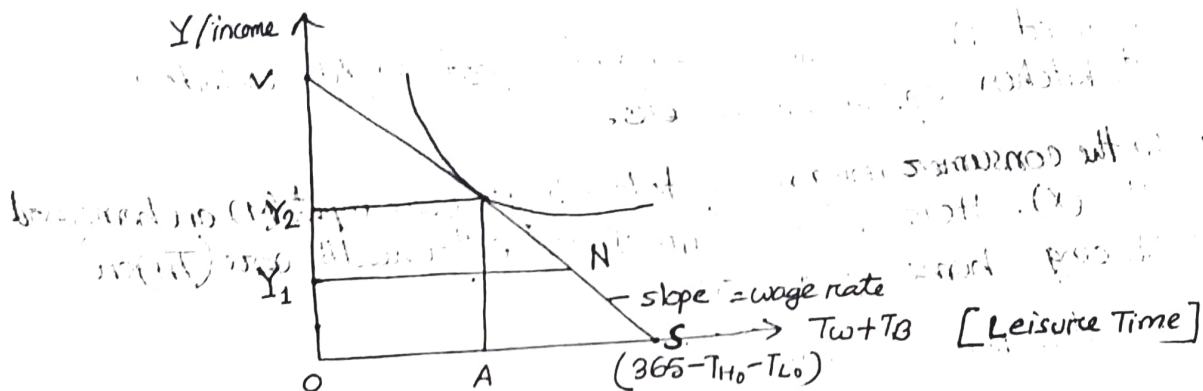


Figure :- Labor - Leisure Trade off.

In this figure, X axis = consumer's work and leisure time
Y axis = income.

Let consumer consider his time spent creating health investment and T_b his leisure time. Further assume that the number of days lost to ill health and the number of days spent on health enhancing activity are fixed.

T_{L0} = Time lost due to illness

T_{H0} = time spent on health activity

The maximum amount of time that he has available to use either work T_w or leisure T_b is thus $365 - T_{H0} - T_{L0}$.

Time available for work/Leisure = $365 - T_{H0} - T_{L0}$

= $T_w + T_b$

The expenditure of time for health producing activities may later improve consumers available, hours of productive activity.

The consumer can increase his utility because of his investment moving from E to E' . This also leads to more future income and may lead to more leisure as well. This improvement in health status also might increase consumers productivity at work, perhaps resulting in a higher wage and a steeper income-leisure curve.

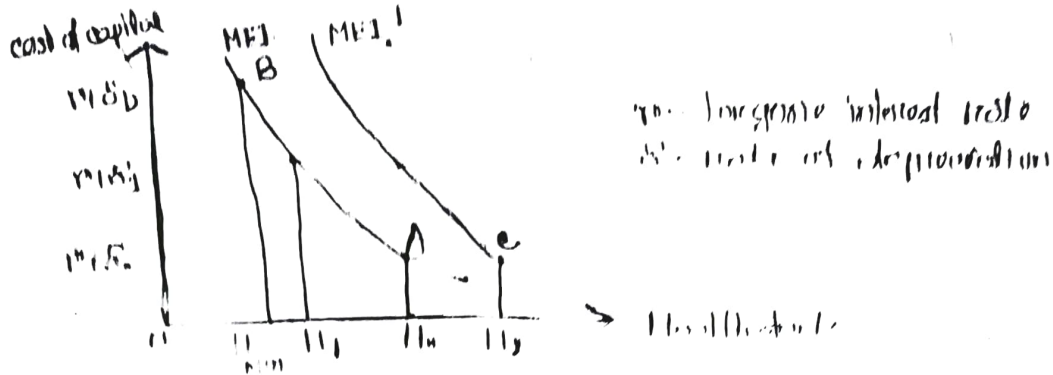
Que 9 Explain in the light of Grossman Model, how the optimal health stock is determined.

MEI: The MEI can be described in terms of the X-Ray machine. Suppose a X-Ray machine's price is 100000 and annual income from it is 20000 that, rate of return is $\frac{20,000}{100,000} = 20\%$. The management would buy this machine if ~~cost~~ rate covers its opportunity cost and depreciation cost.

If management considered to machines then the second machine would have a lower rate of return than the first. Because first machine would assign it to the highest priority uses that is highest rate of return. And second machine would assign it to the lesser priority uses (might be idle on occasionally). The clinic will purchase the second machine only if its rate of return were still higher than interest plus depreciation.

Optimal Health Stock

The MEI determines the pattern of rate of return declining as the amount of health investment increases that is, if other machines probably could be needed added at successively lower rates of return.



In figure 10 stands for long-term interest rate and δ stands for rate of depreciation. In figure 10, equilibrium occurs at point A where the amount of capital at which the marginal efficiency of investment just equals the cost of capital that is $10\% = 10\%$.

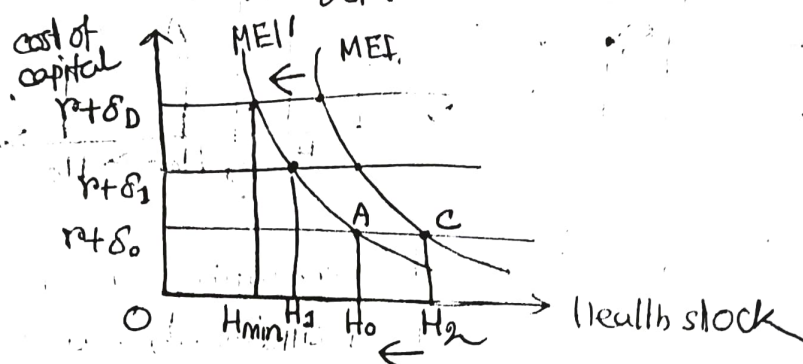
The MEI curve is downward sloping because the production function for healthy days exhibits diminishing marginal returns. The cost of capital for health would similarly reflect the interest rate plus depreciation rate.

Thus the optimal demand for health is likewise given at the intersection of the MEI curve and cost of capital curve ($r + \delta$).

How does a persons investment in health change in response to change in age, wage and education?

Ans:

Age: The health of older people is likely to deteriorate faster than the health of younger people. In our following figure the optimal health at young age H_0 is greater than H_1 the optimal stock at the older age. Higher ~~education~~ depreciation rates increase the cost of holding health capital stock. So people will increase their gross investment in health as their age. That is the elderly would demand more medical care than young. Though the investment on health increases when age increase but the optimal health will decline as the persons age. Because depreciation rate is very high for old people, in this case MEI curve shift to left.



From figure, optimal health stock of younger age H_0 is greater than H_1 optimal stock at the old age. When age increase then depreciation of health stock increase.

Finally health stock investment return decreases for this reason MEI curve shift to the left. The optimal health stock defined H_2 where $H_2 < H_0$ and $H_2 > H_1$ for investment on health stock.

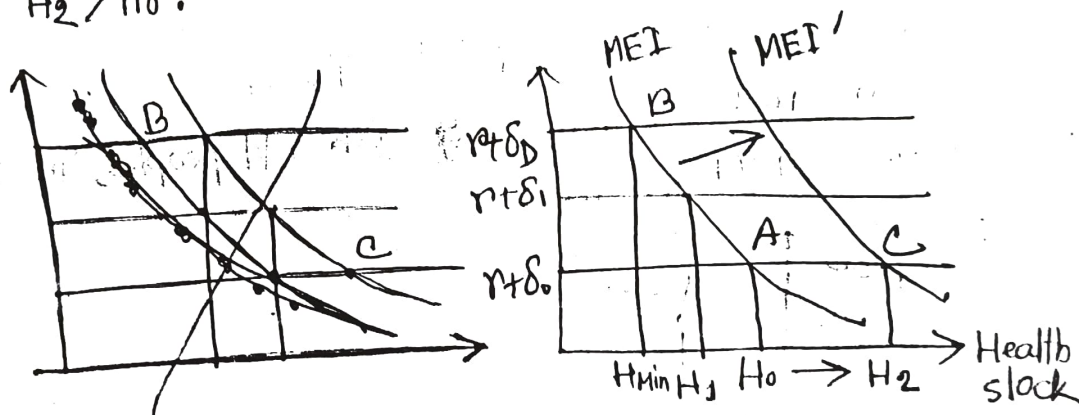
Wage Rate ÷

Increased wage rate increase the return obtained from healthy days. So higher wage imply a higher MEI curve. Here in our ~~curve~~ Figure Higher MEI is MEI' .

MEI = describes the lower wage case.

MEI' = describes the higher wage case.

Now after increasing wage new equilibrium will be at point e' . Higher wage will imply a higher optimal level of health stock H_2 where $H_2 > H_0$.

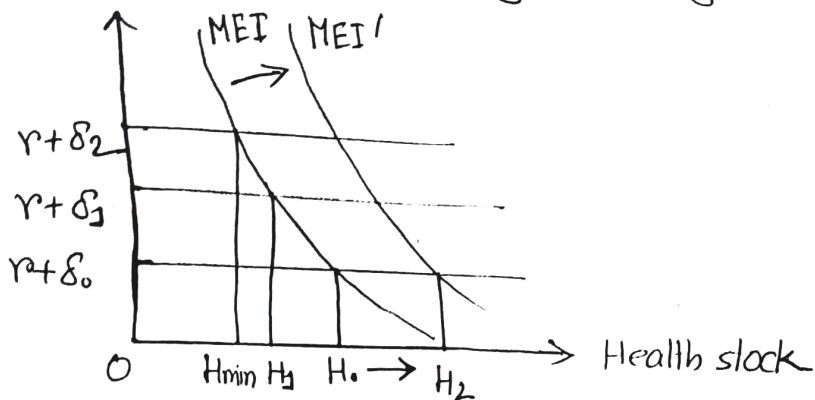


optimal healthy stock would not necessarily drop to H_{min} directly upon ones net retirements but it would do so only when depreciation rate become sufficiently servere.

Education &

Education may 'improve the efficiency with which one can produce investments to health goods next home goods.

Example of improved efficiency may include improved ability to follow instruction regarding medicine or better knowledge on harmful effects of smoking, drinking or addictive drug.



In our above figure MEI curve shows the marginal efficiency of investment for the consumer with a low level of education [measured by years of schooling] while the MEI' curve shows the same person will a higher level of education.

So at last, we can say that more educated people will choose higher optimal health stock H_2 than the less educated H_0 .

In this case investment on health production will increase. Health can be generated at low cost for more highly educated people.